Theorem 1 (Napoleon's Theorem)**.** *Let $\triangle ABC$ be any triangle. Construct equilateral triangles all externally, or all internally, on the three sides of $\triangle ABC$, and let L, M, and N be the centroids of the equilateral triangles built on BC, CA, and AB, respectively. Then $\triangle LMN$ is equilateral.*

Proof. Work in vector coordinates in the plane, and let J denote rotation by 90° counterclockwise. There is a sign $\varepsilon \in \{1, -1\}$ such that

$$L = \frac{B+C}{2} + \varepsilon \frac{\sqrt{3}}{6} J(C-B),$$

$$M = \frac{C+A}{2} + \varepsilon \frac{\sqrt{3}}{6} J(A-C),$$

$$N = \frac{A+B}{2} + \varepsilon \frac{\sqrt{3}}{6} J(B-A).$$

The same sign ε is used for all three sides: one choice corresponds to the external construction, and the other to the internal one.

Subtracting gives

$$M - L = \frac{A-B}{2} + \varepsilon \frac{\sqrt{3}}{6} J(A+B-2C),$$

$$N - M = \frac{B-C}{2} + \varepsilon \frac{\sqrt{3}}{6} J(B+C-2A).$$

Using $J^T J = I$ and $J^2 = -I$, a direct expansion yields

$$|M-L|^2 = |N-M|^2$$

and

$$(M-L) \cdot (N-M) = -\frac{1}{2}|M-L|^2.$$

Thus the two consecutive side vectors have the same length and make an angle of 120°. Equivalently, the interior angle at M is 60° and $LM = MN$.

By the law of cosines,

$$|L-N|^2 = |L-M|^2 + |M-N|^2 - 2|L-M||M-N|\cos 60^\circ = |L-M|^2.$$

So $LN = LM = MN$, and therefore $\triangle LMN$ is equilateral. $\square$